Advanced Topics in Computer Graphics II

Universität Bonn
Institut für Informatik II
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Sheet G03 - Subdivision

Submit solutions for the theoretical and practical part via eCampus by Mo, 02.11.2026, **10:00**.

Practical Part

Please download the zip file `framework_g03.zip` from eCampus and unzip them into the `exercises` folder. Configure the project like last time.
Submit this exercise by uploading the complete source file (source.cpp) onto eCampus!

Assignment 1) Subdivision Surfaces - Sqrt3 *(6Pts)*

In the file `exercises/exercise03Subdivision/source.cpp`, you will find the exercise for this assignment.

- In the function **subdivide_mesh**(...), implement the computation of the new positions of the old vertices.
- In the same function **subdivide_mesh**(...), implement the face splits.
- In the same function **subdivide_mesh**(...), implement the edge flips.

Theoretical Part

Assignment 2) Subdivision Curves *(3 + 1*Pts)*

Let $\mathcal{P} = \left\{(0,0)^\top, (1,0.5)^\top, (1.4,1)^\top, (0.8,1.2)^\top, (0.3,0.8)^\top\right\}$ be a control polygon starting at $(0,0)^\top$ and ending at $(0.3,0.8)^\top$. Subdivide this polygon using

a) Corner-cutting with weights (3/4, 1/4) and (1/4, 3/4).
b) the odd rule (0.5, 0.5) and even rule (1/8, 6/8, 1/8).
c) To what kind of curve (type and degree) does Corner-cutting converge to?

Bonus: Implement both variants (in a language of your choice, e.g. Python) of the subdivision and iterate them until convergence.

Assignment 3) Subdivision Surfaces *(5Pts)*

a) Describe how the three subdivision algorithms **loop/catmull-clark/sqrt3** work. Go through every step and describe how you can subdivide a mesh using the respective approach approach.
b) What are the main differences between the subdivision approaches? Which one would you use in which situation?

Good luck!